

[Dashboard](#) / [My courses](#) / [MA-224-G 24H](#) / [Tests](#)/ [Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)**Started on** Wednesday, 23 October 2024, 1:21 PM**State** Finished**Completed on** Wednesday, 23 October 2024, 1:41 PM**Time taken** 19 mins 54 secs**Marks** 2.75/3.00**Grade** 2.75 out of 3.00 (91.67%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Partially correct

Mark 0.75 out of 1.00

Consider the following regular expression.

$ex = z??X?$

Which of the following regular expressions accepts at least the same words as ex ?

☒ $z?X?$ ☒ $z?X^*$ ☐ $z?X^+$ ☐ $z^*X?$ ☒ z^*X^* ☐ z^*X^+ ☐ $z+X?$ ☐ $z+X^*$ ☐ $z+X^+$ ☐ None of the above

Question 2

Correct

Mark 1.00 out of 1.00

Apply the refinement rule "Merge Assert" (see below) to the following specification statement.

Specification statement: $\{ p \leq y \} ; \{ 0 \geq p \} \ p, k, y := p', k', y' \mid p' \neq y \Rightarrow k' \neq p$

Refinement rule Merge Assert (Mer)

$\{ \text{pre1} \} ; \{ \text{pre2} \}$
 $[=$
 $\{ \text{pre1 and pre2} \}$

Fill in the correctly refined specification statement. You can copy parts of the original statement.

$\{ p \leq y \text{ and } 0 \geq p \} \ p, k, y := p', k', y' \mid p' \neq y \Rightarrow k' \neq p$

Your last answer was interpreted as follows:

$\{ p \leq y \text{ and } 0 \geq p \} \ p, k, y := p', k', y' \mid p' \neq y \Rightarrow k' \neq p$

Question 3

Correct

Mark 1.00 out of 1.00

Consider the following relation R over integers.

$xRy \text{ iff } x + y = 56$

Check the properties of the relation R.

The relation is reflexive:

False

The relation is symmetric:

True

The grammar is antisymmetric:

False

The grammar is transitive:

False

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 314

Signing code:

Your last answer was interpreted as follows:

35287

[◀ Test 1 \(topics 1-3: Introduction, Concepts, Induction, Recursion, Grammars\)](#)

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